Lemma 2.43 ([17, Lemma 9.2])**.** *Let $L_1, L_2 \subseteq \mathrm{VF}^m$ be two minimal limit sets of f at a. Then the closures of L_1, L_2 coincide.*

This lemma justifies writing $\lim_{A\to a} f = L$ when L is a closed (hence the unique) minimal limit set of f at a, in particular, when L is finite and minimal.

Let $A \subseteq \mathrm{VF}^n$ be a definable set with $\dim_{\mathrm{VF}}(A) = n$ and $f : A \longrightarrow \mathcal{P}(\mathrm{VF}^m)$ a definable finitary function. We say that f is *continuous* if $\lim_{A\to a} f = f(a)$ for all $a \in A$.

Unapplied R039 proposal and arity question for author review. Keep the finite RV coding and check all labels simultaneously on a neighborhood where the label set is constant. The empty-label case is the constant empty correspondence.
Additional arity issue. If $n = 0$, then $\mathrm{VF}^0 = \{e\}$. For $A = \{e\}$ and $f(e) = \{0\}$, the puncture $A \setminus \{e\}$ is empty. Hence the empty set is a limit set, and $\{0\}$ is not minimal: this constant correspondence is nowhere continuous under the stated definition. Since $\dim_{\mathrm{VF}}(A) = 0$, it cannot be discarded as a smaller-dimensional exception. The draft therefore proposes $n > 0$ in Lemma 2.44, while preserving the punctured limit convention. Its curve-selection use is unary. At the later fiberwise-boundary use, the assumed disc in S_1 already forces positive source dimension. Both the proof expansion and the arity restriction await approval.

Lemma 2.44. *Suppose $n > 0$. Then f is continuous almost everywhere.*

Proof. This can be reduced to the familiar case that f is indeed a function into VF^m as follows. By Lemma 2.12 and compactness, there are a definable set U in RV and a definable function $g : A \times \mathrm{VF}^m \longrightarrow A \times U$ such that, for all $a \in A$, $\mathrm{pr}_A(g(a, b)) = a$ and $g \restriction (a \times f(a))$ is injective. For each $t \in U$, let

$$A_t = \{a \in A \mid (a, t) \in g(a \times f(a))\}$$

and $f_t \subseteq f$ be the obvious t-definable function $A_t \longrightarrow \mathrm{VF}^m$. The discontinuity locus E of f is definable: the limit condition is first-order, and minimality of the finite set $f(a)$ can be tested by removing one of its elements at a time. Suppose for contradiction that $\dim_{\mathrm{VF}}(E) = n$. By Lemma 2.17, E contains a nonempty open polydisc. Apply generic local constancy [17, Lemma 8.9] to the finite label-set map

$$a \longmapsto g(a \times f(a))_{\mathrm{RV}}.$$

After shrinking to a nonempty open polydisc $P \subseteq E$, this set is a fixed finite set T. If T is empty, then f is the constant empty correspondence on P and is continuous there, a contradiction.

For each $t \in T$, the inverse branch f_t is defined on all of P. Extend it by zero outside A_t and apply Lemma 2.36. The finitely many branch-exception sets have union of VF-dimension less than n, so choose $a \in P$ outside their union. All branches are continuous at a. Their finite union has limit set $f(a)$; this limit set is minimal because the distinct values at a can be separated by sufficiently small discs and, since $n > 0$, the punctured polydisc still accumulates at a. Thus f is continuous at a, contradicting $P \subseteq E$. Hence $\dim_{\mathrm{VF}}(E) < n$. □

Referee R039 [Gap; revised] – One continuous branch is not enough.
Original source line 1068. The argument finds one t for which A_t has full dimension and f_t is generically continuous. That alone does not account for all branches of f. Apply generic local constancy to the finite RV-label set $a \mapsto g(a \times f(a))_{\mathrm{RV}}$, then simultaneously apply generic continuity to its finitely many inverse branches on each neighborhood. To justify the global dimension conclusion, suppose the discontinuity locus had full dimension and choose an open polydisc in it where this label set is constant; the finite collection of branch-exception sets cannot fill that polydisc. The empty label set simply gives the constant empty correspondence and should be handled separately. This supplies the missing coverage argument.